



Public Transit User Dynamics During Boarding and Alighting: A Typology-Based Framework for Perceived Crowding and Trajectory Simulation Under Crowded Conditions in [location] Context

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Things I want to share today

Background study

General problem

Rationale

Research objective

Research Question

Future task

General Problem

- Emerging cities have been increasingly promoting the use of public transportation.
- This trend has led to a growing demand for public transport systems, particularly metro rail and city bus services, within urban contexts.
- Instead of non-peak hour, passengers demand intensifies during peak hour serving various trip purposes such as commuting to work and school, hence, maintaining reliable and efficient operations becomes increasingly challenging.
- Among the various operational processes in public transport, the boarding and alighting process at stops and stations is one of the most critical determinants of service efficiency, which could govern dwell time (Thoreau et al, 2016).
- Additionally, Daamen et al. (2008) noted that dwell time is mostly consumed by the boarding and alighting process itself, identifying it as the primary driver of stop duration.

General Problem

The direct impact of boarding and alighting process on public transport services and reliability

- Previous investigated that boarding and alighting time could be factor directly linked to operational problem, i.e., **bus bunching**, by simulating with cutting just one second per passenger in boarding and alighting time is sufficient to visibly reduce bunching, even without any other operational change (Enayatollahi et al. 2019).
- Under crowded conditions when in-carriage density exceeds a critical threshold (LOS D, 2.97 persons/m²) → alighting time: more than doubles (+126.3%); boarding time: more than doubles (+111.8%) (Shen et al. 2026)

Passenger dynamics in boarding and alighting process

- Passengers' dynamics affect boarding and alighting time.
- Crowded condition introduces a dimension of complexity that extends beyond actively moving passengers alone.
- Under high occupancy, passengers remaining stationary, whether inside the vehicle or waiting on the platform, are no longer passive elements of the environment. They become active participants whose spatial positioning, yielding decisions, and behavioral responses directly condition the movement of alighting and boarding passengers around them.
- Shen et al. (2026) provided simulation study which included carriage standees and platform-waiting pedestrians as individual agents.

Rationale

The need on passenger typology consideration

- Most prior studies have primarily focused on passenger movements associated with entering and exiting vehicles during boarding and alighting processes, meaning that passengers who are actively moving are included in the analysis process (Qu Y., et.al., 2019; Seriani and Fernandez, 2015).
- When crowded situations happen, stationary passengers are also active participants whose spatial adjustments and yielding decisions directly shape the flow around stationary passengers.
- Besides, even within the subset of actively moving passengers, behavioral profiles differ systematically between types.

Empirical findings on behavioral profiles differ among moving passengers

- As platform and vehicle occupancy rises during peak hours, time pressure intensifies, pushing passengers from conservative, rule-following behavior toward more competitive behavior that can escalate into door conflicts (Qu et al., 2019).
- This pressure is asymmetric: missing the door costs a boarding passenger only a wait for the next train but costs an alighting passenger a trip to the next station and back, driving alighting passengers to move faster and more consistently, while boarding passengers behave more variably and hesitantly (Qu et al., 2019).
- This asymmetry is not unique to metro contexts: smart-card data from Singapore's bus network shows higher on-board occupancy increases boarding time per passenger (+0.340 s/pax) through in-vehicle friction, while decreasing alighting time per passenger (−0.082 s/pax), as crowding discomfort pushes passengers to exit faster (Sun et al., 2014)

If behavioral distinctness is already observable between alighting and boarding passengers alone, the case for distinguishing passenger types in simulation models becomes empirically unavoidable

Rationale

Previous studies on passenger type distinction

Li Z. et al. (2020) made an early attempt, classifying passengers into **alighting, boarding, and staying (inside metro)** types in their SFM simulation, giving staying passengers a small reactive capacity to shift position when moving passengers come nearby, a step beyond treating them as fixed obstacles.

Yining Jia et al. (2026) explicitly categorizes metro passengers into **boarding, alighting, and remaining on-board groups**, but situates this within normal operating conditions where remaining passengers are typically seated, an assumption supported by controlled experiments (Fu L. et al., 2023) but one that collapses under crowded conditions, where remaining passengers are predominantly standing, spatially constrained, and compelled to actively respond to surrounding movement.

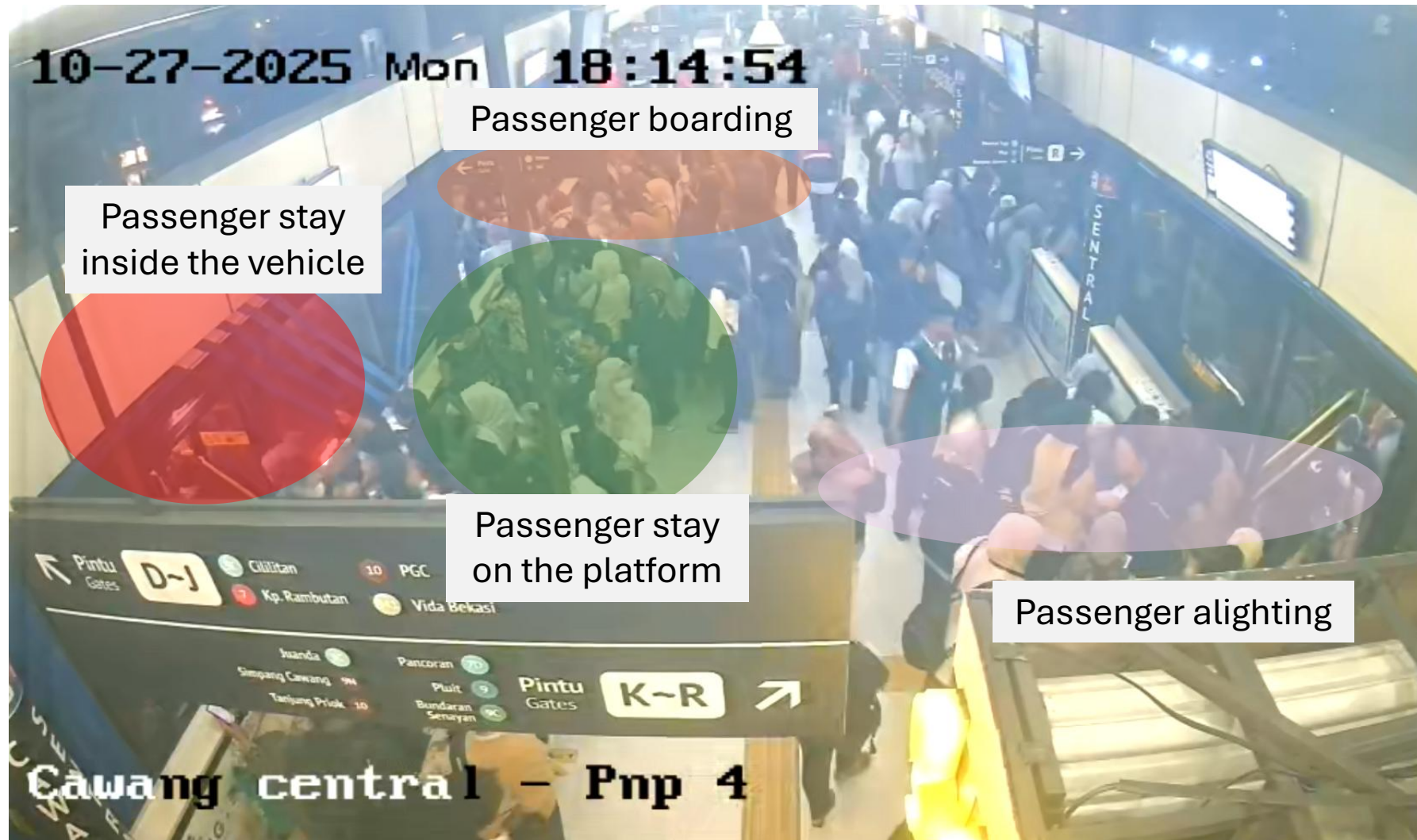
Shen et al. (2026) advances further with four passenger types: alighting, boarding, carriage standees, and platform-waiting pedestrians, matching the full range this study proposes, but treats stationary passengers only as physical objects with zero speed that block or deflect movement through force equations, without any capacity to decide or respond to the surrounding situation.

Consequently, even though recent studies have begun to include all four passenger types, stationary passengers are still only represented physically (as an obstacles) rather than as agents who decide when to yield, how to move, and how to react as crowding around them increases

Rationale

- To address this gap, this study proposes four passenger types defined by their state during the boarding and alighting process:

State	Type	Represented by
Non-stationary	Passenger alighting	PA_m
	Passenger boarding	PB_m
Stationary	Passenger stay inside the car	PC_s
	Passenger stay on the platform waiting for boarding process	PP_s





Rationale – perceived crowding

- According to Shen et.al, (2026) result, when more passengers alight at once, the alighting process/exit time (measured from the moment the train doors open passenger passing through the platform until that passenger reaches their chosen exit) finishes faster → the fastest exit time of alighting passenger drops from 23.15 s when 100 alighting passengers, to 16.40 s with 500 alighting passengers.
- From operational point of view, this result looks good.
- But the result cannot tell us what this costs the passengers waiting on the platform, since in that model waiting passengers were set to be zero speed and no ability to perceive, decide, or react (**note**: using SFM, so they are represented only as physical objects to be displaced. Any cost borne by these passengers is invisible to the model by construction, not because the cost is small, but because nothing in the model is capable of registering it.
- Is this a serious problem, or an acceptable simplification? That depends on one thing: whether being pushed through actually costs the waiting passenger anything.

Rationale – perceived crowding

- Stokols (1972) introduced theoretical concept on distinction between crowding and density: density is a physical condition, while crowding is the psychological experience that arises when a person feels the space available to them has fallen below what they need. The two are not the same, and it is the experience, not the density, that drives how a person behaves. Something happens inside the passenger, over and above the physical displacement.
- Kalb and Keating (1981) show that crowding, as an experience, has to be measured on its own terms: **feeling crowded** and **judging an environment to be crowded** are distinct constructs, and only feeling crowded is associated with hindrance, restriction, and confinement.
- Crowding, as an experience, is therefore not discomfort at the presence of others. It is the experience of constraint.
- The subjective state is separate, and it is the one with consequences.
- In a transit setting, Mahudin et al. (2012) found that this feeling appears when passengers sense their personal space has been exceeded, and that it brings frustration, stress, and a sense of being hindered. Two passengers standing in identical density can therefore be having entirely different experiences and behaving accordingly.

Rationale – perceived crowding

- The cost (feeling crowded) is there.
- When it comes Stokols (1972) identifies the natural response to feeling crowded: people leave. They withdraw from the crowded space to regain room.
- This is where the boarding and alighting process becomes distinctive, because for two of the four passenger types, this response is unavailable.
- A passenger waiting on the platform PP_s is typically committed to boarding at that door.
- A passenger standing inside the carriage PC_s is enclosed and cannot leave at all.
- Both accumulate the experience of crowding, and neither can discharge it in the way Stokols (1972) describes.
- If they cannot respond by leaving the situation, then whatever response they make has to happen inside the situation, in the small repertoire of actions still open to them. For PC_s : whether to yield or resist. For PP_s becoming PB_m : whether to wait or push forward
- As Qu et al. (2019) proposed that a passenger who has queued a long time, or repeatedly made way for others, builds up time pressure → shapes how they board (shifting passenger from conservative behavior to more forceful behavior).

Rationale – perceived crowding

- Given the Shen et al. (2026) result, it could raise: does a faster exit time represent a genuine improvement to the process, or does it simply shift the burden onto passengers?
- It requires knowing how each type of passenger actually perceives crowding, and how that perception governs what they do.

Rationale – perceived crowding

Passenger Type	State	Governing attitudinal / perceptual construct	Behavioral response under crowded conditions
PA_m	Non-stationary — alighting phase	Perceived crowding while forcing a path through a dense platform; time pressure from the heavier penalty of missing the door; tolerance for breaking the alight-first rule.	Moves urgently and assertively through the door; navigates around PC_s inside the carriage and PP_s on the platform. Aligned motives produce consistent, low-variance movement
PP_s	Stationary — alighting phase	Perceived crowding; withdrawal unavailable, typically being committed to board at that door	Yielding decisions and spatial repositioning as PA_m approaches and passes through on the platform area

Rationale – perceived crowding

Passenger Type	State	Governing attitudinal / perceptual construct	Behavioral response under crowded conditions
PB_m	Non-stationary — boarding phase	Perceived crowding while moving into higher density, pulling against the goal of boarding; compliance with the alight-first rule; crowding and pressure carried over from the PP_s state. Withdrawal available in principle but costly and goal-conflicting	Waits for PA_m to clear, or pushes forward early; under high accumulated pressure, compliance weakens and door conflict increases
PC_s	Stationary — throughout both phases	Perceived crowding discomfort; withdrawal unavailable; threshold for giving way	Repositions to make room as PA_m passes and as PB_m enters; willingness to give way shaped by accumulated discomfort, not by physical force alone

Rationale – heterogenous passengers dynamics simulation in continuous space

Under development

Research Objective

To develop typology-informed understanding of passenger behavioral dynamics during the boarding and alighting process under crowded public transport conditions, by integrating perceived crowding as a subjective behavioral alongside simulation framework that accounts for heterogeneous nature of passenger states.

Research Question

RQ1 – Building upon rationale – perceived crowding

How does perceived crowding influence the behavioral responses of alighting (PA_m), boarding (PB_m), carriage-staying (PC_s), and platform-waiting (PP_s) passengers during the boarding and alighting process under crowded conditions?

RQ2 – Building upon rationale – heterogeneous passengers dynamics simulation in continuous space